COURSE-BASED PROJECT DOCUMENTATION | DOC-18

Project Conclusion & Academic Review

Summary of Achievements, Compiler Concepts Demonstrated, & Final Remarks

1. Summary of Project Achievements

The Java 21 Syntax Checker project successfully constructed a high-precision, robust compiler front-end tool targeting the finalized Java 21 LTS language specification. The project achieved:

- ? A fully compliant GNU Flex 2.6.x lexical DFA scanner with location tracking.
- ? A deterministic GNU Bison 3.8.x LALR(1) shift-reduce parser resolving all Java 21 ambiguities.
- ? A dual-target C native engine compiled into javacheck.dll and javacheck.exe.
- ? A modern Python 3.10+ Tkinter GUI with in-memory ctypes FFI, Command Palette, and error highlighting.
- ? A verified 100% pass rate across 59 parser test cases and 14 GUI integration tests.

2. Core Compiler Design Concepts Demonstrated

Theoretical Concept	Practical Implementation in This Project
Regular Expressions & DFAs	Lexer token specification (java_lexer.l) using Flex state machine
Context-Free Grammars	Formal Java 21 grammar rules in Bison (java_parser.y)
LALR(1) Parsing Tables	Shift-reduce state transitions over 998+ parser states
Ambiguity Resolution	%nonassoc precedence rules resolving dangling-else & switch gu...
Location Tracking	YY_USER_ACTION synchronizing yylloc coordinates with yylineno
Native FFI Interoperability	Direct C ABI shared library loading via Python ctypes

3. Concluding Statement

By successfully bridging classical compiler construction theory with modern programming language syntax and desktop tool design, this project fulfills all requirements of the Course-Based Project with academic rigor, technical excellence, and practical usability.